

YekBûn — Device Control Mobile APIs

Handoff for mobile developers · Auth not required · Base: <https://api.appdash.yekbun.org/api> · Cache pack v1.15.0

Boot flow (important)

- 1) App start → **GET /api/app/device-profile** with device hardware
- 2) Apply **data.runtime** + **data.cache** locally (video max_active, reels initial, cache MB budgets...)
- 3) Periodically **POST /api/app/device-telemetry**
- 4) On crash/ANR → **POST /api/app/device-crash**

No separate cache endpoint — cache is inside resolve response.

Endpoints

Method	Path	Purpose
GET / POST	/api/app/device-profile	Resolve device + runtime + cache
POST	/api/app/device-telemetry	Heartbeat upsert by device_id
POST	/api/app/device-crash	Crash / ANR → problem group

1. Resolve Profile (start API)

GET or POST /api/app/device-profile

Params: **ram** ("4 GB") or **ram_class** (4|6|8|12+), **cpu_tier** (low|mid|high|flagship|entry), **os**, **os_version**, **manufacturer**, **model**.

Request

```
GET /api/app/device-profile?ram=4%20GB&cpu_tier=low&os=Android&os_version=13
```

Response (trimmed — Low tier example, cache v1.15.0)

```
{
  "success": true,
  "message": "Device profile resolved.",
  "data": {
    "matched_by": { "ram_class": "4", "cpu_tier": "low", "platform": "android" },
    "profile": {
      "key": "low", "name": "Low", "version": "v1.14.1",
      "cache_profile_key": "low", "runtime_profile_key": "low",
      "memory": { "max": 900, "warn": 700, "critical": 820 }
    },
    "runtime": {
      "key": "low",
      "api": { "max_parallel": 3, "retry": "smart", "timeout_ms": 15000 },
      "feed": { "batch_size": 8, "strategy": "lazy" },
      "video": {
        "autoplay": true, "quality": "480p", "max_active": 2,
        "preload": 4, "buffer": 8, "decoder": "hardware", "memory_budget": 384
      },
      "reels": { "initial": 1, "next_preload": 1, "video_queue": 4, "strategy": "lazy" },
      "rendering": { "image_quality": "medium", "strategy": "minimal" },
      "network": { "mode": "conservative", "bandwidth_saver": true }
    },
    "cache": {
      "key": "low", "version": "v1.15.0",
      "allocation": { "total_size": 266, "max_size": 319, "mode": "hybrid", "expandable": true },
      "categories": [
        { "id": "feed", "max_size": 32, "priority": "high", "auto_cleanup": true },
        { "id": "video", "max_size": 48, "priority": "high", "auto_cleanup": true },
        { "id": "reels", "max_size": 36, "priority": "high", "auto_cleanup": true }
      ],
      "cleanup": { "mode": "automatic", "order": "priority",
        "protected_categories": ["system","fonts","languages","policy"] },
      "sync": { "on_app_start": true, "wifi_only": false, "mobile_data": true, "max_sync_size": 106 }
    }
  }
}
```

Cache budgets by RAM tier (v1.15.0)

Tier	RAM	Total	Video	Reels	Feed	Notes
Entry	≤4 GB	169 MB	28	20	20	wifi-only sync; maps off
Low	4 GB	266 MB	48	36	32	safe feed scroll
Balanced	6–8 GB	476 MB	96	64	48	default
High	8–12 GB	784 MB	180	120	80	hybrid cleanup
Ultra	12 GB+	1248 MB	320	200	120	prefetch-friendly

Apply category **max_size** as disk cache caps. Respect **cleanup** + **protected_categories**. Entry: no background sync, **wifi_only=true**.

Video / Reels — what mobile must apply from runtime

Field	Meaning
video.max_active	How many videos may play at once
video.quality / preload / buffer	Playback quality + preload/buffer seconds
video.autoplay	Allow autoplay or not (Entry=false)
reels.initial / next_preload	First reels load + next prepare count
reels.video_queue / strategy	Queue size; prefer lazy on Entry/Low

2. Telemetry Heartbeat

POST /api/app/device-telemetry · Required: **device_id**

```
{
  "device_id": "D-abc123",
  "name": "Galaxy A14", "model": "SM-A145F", "manufacturer": "Samsung",
  "os": "Android", "os_version": "13", "ram": "4 GB", "cpu_tier": "Low",
  "profile_key": "low",
  "cache_used_pct": 84, "memory_usage_pct": 92, "fps": 32,
  "health_score": 42, "crash": "None", "crash_count": 0,
  "status": "healthy", "app_version": "3.14.2", "app_version_bucket": "latest"
}
```

3. Crash / ANR Report

POST /api/app/device-crash · Required: **device_id**, **problem_type**

```
{
  "device_id": "D-abc123",
  "problem_type": "OutOfMemory",
  "crash_cause": "OutOfMemory in FeedRenderer.mountItem()",
  "crash_signature": "OOM · FeedRenderer#mountItem",
  "affected_screen": "Feed",
  "profile_key": "low",
  "os": "Android", "os_version": "13", "ram": "4 GB", "cpu_tier": "Low",
  "manufacturer": "Samsung", "model": "SM-A145F",
  "app_version": "3.14.2", "memory_at_crash": 96, "severity": "Critical"
}
```

MongoDB collections (for QA / backend check)

API	Writes?	Collection
device-profile	No (read)	device_profiles + runtime_profiles + cache_profiles
device-telemetry	Yes	device_telemetry
device-crash	Yes	problem_devices (+ bumps device_telemetry)

Notes

- Auth not required for these three endpoints.
- Content-Type: application/json for POST.
- Seeded tiers: Entry / Low / Balanced / High / Ultra.
- If resolve returns 404 → ask backend to run: `php artisan device-control:seed-defaults`